Limits of Sequences: Practice Problems

Part A: L'Hôpital's Rule

For each sequence, determine whether it converges, and if so find its limit. Treat $a_n = f(n)$ as the values of a function $f(x)$ and apply L'Hôpital's Rule to $\lim_{x\to+\infty} f(x)$.

1. $\left\{\dfrac{\ln n}{\sqrt{n}}\right\}_{n=1}^{+\infty}$

2. $\left\{\dfrac{n^2}{e^n}\right\}_{n=1}^{+\infty}$

3. $\left\{\left(1+\dfrac{3}{n}\right)^n\right\}_{n=1}^{+\infty}$ *(Hint: let* $y = \left(1+\frac{3}{x}\right)^x$ *and first find* $\lim_{x\to+\infty} \ln y$*.)*

Part B: The Squeezing Theorem

Use numerical evidence to make a conjecture about the limit, then use the Squeezing Theorem for Sequences to confirm that your conjecture is correct.

4. $\displaystyle\lim_{n\to+\infty} \frac{\cos^2 n}{n}$

5. $\displaystyle\lim_{n\to+\infty} \frac{n+\sin n}{2n+1}$

6. $\displaystyle\lim_{n\to+\infty} \left(3^n+5^n\right)^{1/n}$ *(Hint: compare* 3^n+5^n *with* 5^n *and with* $2\cdot 5^n$*.)*

Part C: Limits of Recursively Defined Sequences

Assume each sequence converges. Use the method of Example 10 (let $a_n \to L$, so $a_{n+1} \to L$ as well, and take the limit of both sides of the recursion) to find the limit.

7. Consider the sequence

$$a_1 = \sqrt{2}, \quad a_2 = \sqrt{2+\sqrt{2}}, \quad a_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}, \quad ...$$

   a. Find a recursion formula for a_{n+1}.

   b. Assuming that the sequence converges, find its limit.

8. $a_1 = 1, \qquad a_{n+1} = \dfrac{1}{2}\left(a_n + \dfrac{5}{a_n}\right)$

9. $a_1 = 1, \qquad a_{n+1} = 1 + \dfrac{1}{a_n}$ *(Write out the first five terms. Do you recognize the numerators and denominators?)*